CARL: Calibration Aware Residual Learning

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Part I

Foundations

0.1 Introduction

Motivation

Forecast uncertainty depends heavily on residual behavior.

Diffusion models can learn complex residual distributions.

Existing diffusion methods do not optimize for calibration.

Calibration is essential for trustworthy uncertainty quantification.

A calibration-aware training paradigm may improve uncertainty quality.

Contribution

Propose Calibration-Aware Diffusion Residual Learning.

Incorporate calibration into diffusion training.

Compare calibration-aware and standard diffusion learning.

Evaluate performance under controlled innovation misspecification.

Demonstrate effectiveness across both classical and deep forecasting systems.

0.2 Literature Review

Conformal prediction, calibration, calibration aware learning, probabilistic forecasting, diffusion models

0.3 Background

0.3.1 Forecasting Models

VAR

RNN

0.3.2 Residual Processes

Residuals as empirical innovation proxies

0.3.3 Diffusion Models

DDPM

Score-based Models

SDE Interpretation

0.3.4 Calibration

Coverage

Reliability

PIT and ECE

0.3.5 Proper Scoring Rules

CRPS

Energy Score

Log Predictive Density

0.3.6 Uncertainty Quantification

Predictive distributions

Forecast trajectories

probabilistic forecasting

Part II

Calibration Aware Diffusion Residual Learning

0.4 Problem Formulation

0.4.1 Forecasting System

Given:

$\hat{y}_t = f(y_{1:t-1})$

compute residuals: $r_t = y_t - \hat{y}_t$ and learn: $p(r)$ using diffusion models.

0.4.2 Standard Residual Learning

Current diffusion models optimize: $\mathcal{L}_{\text{diffusion}}$ through score matching or denoising.

Calibration is evaluated only after training.

0.4.3 Proposed Framework

Calibration becomes part of the optimization objective: $\mathcal{L} = \mathcal{L}_{\text{diffusion}} + \lambda \mathcal{L}_{\text{calibration}}$

0.5 Calibration Objectives

0.5.1 Coverage Loss

Penalize deviations from nominal coverage:

$\mathcal{L}_{\text{coverage}} = (\hat{C} - (1 - \alpha))^2$

0.5.2 PIT-Based Calibration Loss

Encourage: $u_i = F_i(y_i)$ to approach Uniform(0,1).

0.5.3 ECE-Based Loss

Penalize confidence-frequency mismatch.

0.5.4 CRPS-Based Loss

Use proper scoring rules as calibration objectives.

0.5.5 Combined Objective

Investigate: $\mathcal{L} = \mathcal{L}_{\text{diffusion}} + \lambda \mathcal{L}_{\text{calibration}}$ for different calibration choices.

0.6 CADRL Training Procedure

Step 1: Fit forecasting model.

Step 2: Extract residuals.

Step 3: Train diffusion residual model.

Step 4: Generate predictive residual samples.

Step 5: Compute predictive distributions.

Step 6: Evaluate calibration.

Step 7: Backpropagate calibration loss.

Step 8: Update diffusion model.

Part III

Forecasting Testbed

0.7 Forecast Models

0.7.1 Classical Benchmark

VAR

Motivation: interpretable, linear, statistical baseline

0.7.2 Deep Benchmark

RNN / DeepAR

Motivation: nonlinear forecasting, representation learning, modern forecasting baseline

Evaluate whether calibration-aware residual learning is effective across fundamentally different forecasting architectures.

Part IV

Numerical Experiments

0.8 Controlled Innovation Misspecification

Determine when calibration-aware diffusion learning improves uncertainty quantification.

DGP 1: Gaussian Innovations $u_t \sim N(0, \Sigma)$ Baseline.

DGP 2: Moderate Heavy Tails $u_t \sim t_5$

DGP 3: Extreme Heavy Tails $u_t \sim t_3$

DGP 4: Mixture Innovations $u_t \sim 0.8N(0,1) + 0.2N(0,9)$

DGP 5: Heteroskedastic Innovations $u_t \sim N(0, \sigma_t^2)$

Experimental Comparisons

Forecast Models: VAR, RNN

Residual Models: Gaussian, Student-t, Bootstrap, Standard Diffusion, Calibration-Aware Diffusion

Training Schemes

Standard $\mathcal{L} = \mathcal{L}_{\text{diffusion}}$ Proposed $\mathcal{L} = \mathcal{L}_{\text{diffusion}} + \lambda \mathcal{L}_{\text{calibration}}$

Part V

Evaluation

0.9 Calibration Evaluation

Coverage: Empirical Coverage, Coverage Error

Reliability: ECE, Reliability Diagrams, PIT Histograms

Probabilistic Forecast Quality: CRPS, Energy Score, Log Predictive Density

Sharpness: Average Interval Width, Sharpness Score

0.10 Robustness Analysis

Distributional Perturbations

Investigate: variance inflation, tail amplification, mixture contamination, regime shifts

Metrics, calibration degradation, coverage degradation, sharpness degradation

Part VI

Real Data Case Study

0.11 Empirical Application

Candidate Domain: Financial returns